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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	HamJinBengs
Selected Domain	2
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth
Project / Solution Name	Gantian (On-leave employee scheduler)
Name (Leader)	Tay Kyhenn
Email & Phone (Leader)	DSC2409061@xmu.edu.my , +6012-5083313
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

SME business owners often waste time having to schedule their employee's work schedules in spreadsheets. For example, restaurant owners have many employees, each working their own shifts, as well as intended roles which they need to keep track of.

Unfortunately, this work is further complicated when employees need to take emergency leaves, which is not uncommon at all. In order to fill in this gap, managers need to spend lots of time going through potential employees with the right availability, role suitability, work-life balance, among many others. This makes it difficult to find suitable substitutes, resulting in an immense amount of time wasted, not to mention the risk of miscommunication.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

Gantian is an intelligent roster scheduling and shift-coverage platform built to eliminate the headache of manual scheduling conflicts and last-minute staffing shortages. It combines shift templates, automated roster generation, and real-time shift coverage into a centralized manager dashboard and a self-service staff portal.

When an employee's leave request is approved, the system automates the entire absence recovery workflow. It instantly flags impacted shifts, opens "coverage cases," and generates a ranked list of eligible candidates. Managers can then select preferred replacements to dispatch shift offers, ensuring no schedule conflicts. To tackle complex scheduling logic, the application uses constraint-satisfaction and heuristic optimization AI:

Automated Constraint Evaluation – a rule engine validates every roster assignment against some hard operational constraints (role match, opening hours, approved leave, shift overlaps, and daily/weekly hour and day limits) and soft fairness constraints (minimum rest hours between shifts, pending leave, and 30-day cover frequency).

Fair Roster Generation – a greedy optimization algorithm fills weekly shift demands by prioritizing zero soft-rule penalties, lowest proportional weekly hour utilization, and fewest days worked, while outputting reasons whenever a staffing gap occurs.

Replacement Candidate Scoring – for coverage, the engine filters out unavailable staff and ranks eligible replacements candidates on a 0–100 score based on remaining weekly capacity, rest duration, and recent cover load.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

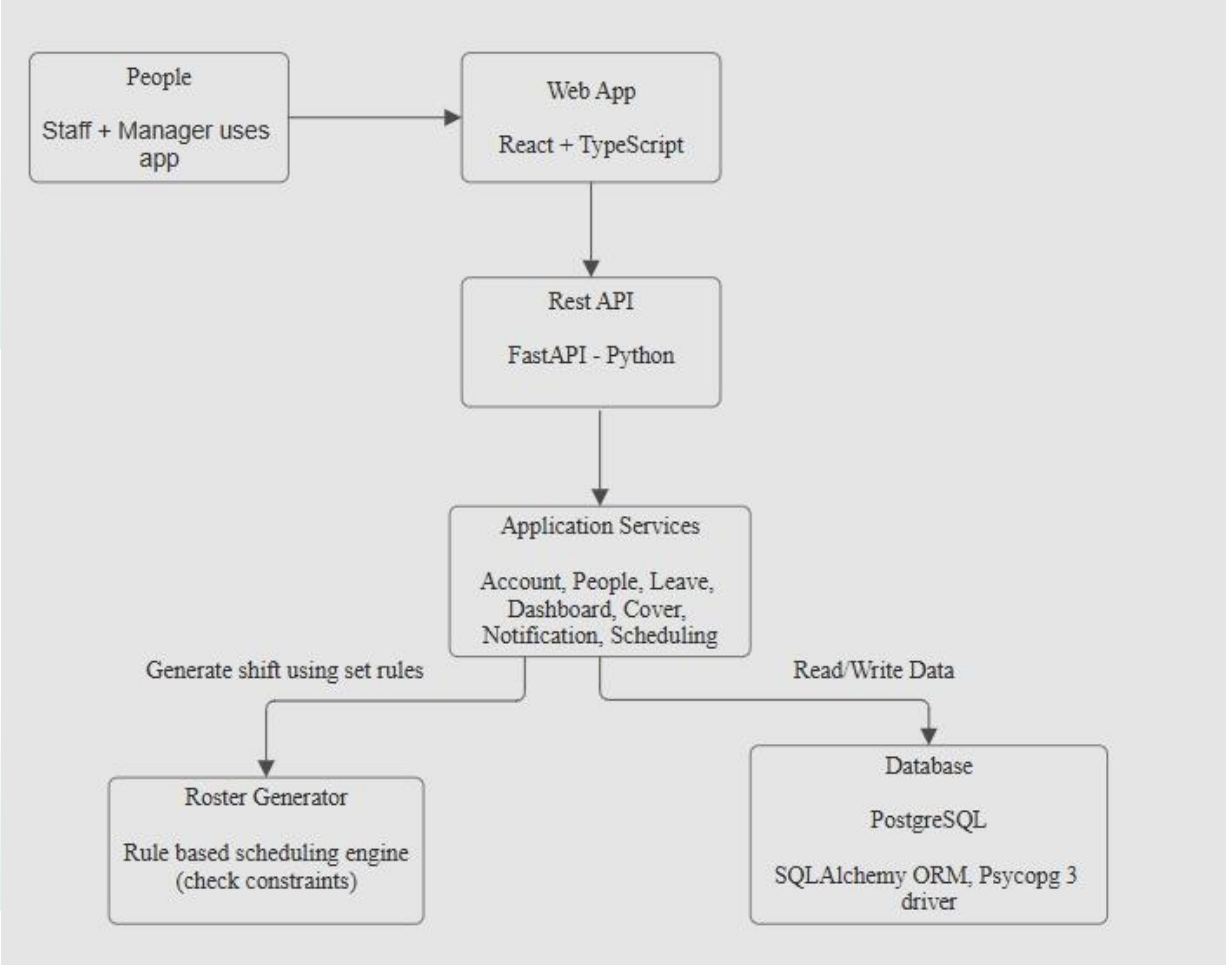
Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 19, TypeScript, Vite, Tailwind CSS v4, Radix UI primitives, Lucide React icons, Wouter, TanStack React Query v5, React Hook Form, Recharts.

Backend: Python with FastAPI, served by Uvicorn. The application is organized into modules for accounts, people, scheduling, leave, cover, notifications, and dashboard.

Application API: An internal REST/JSON API under OpenAPI 3.1. The frontend uses a generated React Query client; Orval generates the client and Zod schemas from the OpenAPI specification.

Database: PostgreSQL, accessed through SQLAlchemy ORM and the Psycopg 3 driver.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Gantian was designed primarily with the market's needs in mind. Gantian strives to deliver customer value to SMEs, and to effectively get rid of customer pains and market gaps which are desperately need filling. Managers always have to deal with the burden of ensuring their companies operate smoothly – saving wasted time in having to keep track of employee schedules can be a lifesaver to spare managers from anxiety!

Gantian was also designed in way to ensure the UI is easily understood. Gantian divides each functionality into their respective "queue", "leave", "roster" etc., separate tabs. Our design focuses on ease of use and being easy to navigate, and we have also patched numerous bugs to eliminate potential friction points.

Additionally, Gantian has plans to be rolled-out in multiple stages, starting from small test groups and gradually gathering feedback from real-world customer use. Real deployment can then be considered once good metrics are obtained in practice, such as low user frustration and high new feature adoption rates.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	8.2 Diversify, innovate and upgrade for economic productivity; 8.8 Protect labour rights and promote safe working environments
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	- Managers will spend up to 6 hours every week on manual roster adjustments. - Manager waste an average of 45-60 minutes per last-minute leave request manually checking spreadsheets and messaging staff on other platform, resulting in a 25% delayed shift-cover rate
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	- % employees with equal workload distributed - % reduction in both scheduling and rescheduling times for managers - % increase in operational efficiency
Phase 2 Impact Target	- Times for manager to schedule suitable replacements to be reduced by at least **50%**

Metric Category	Your Response
Your realistic, percentage-based target by the final prototype presentation *e.g., "20% improvement in computational resource use."*	- Scheduling conflicts to be eliminated entirely (**100%**) - Operational efficiency to be increased by **20%**

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

N/A

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React and React DOM	**Code — frontend UI framework**	**MIT**
FastAPI	**Code — backend API framework**	**MIT**
Pydantic	**Code — data validation**	**MIT**
SQLAlchemy	**Code — database ORM**	**MIT**
PostgreSQL	**Code — database system**	**PostgreSQL License**
Radix UI	**Code — accessible UI components**	**MIT**
TanStack React Query	**Code — API data fetching and caching**	**MIT**
TypeScript	**Code — JavaScript type checking**	**Apache-2.0**

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

N/A

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Signature: Tay Kyhenn

Date: 25.10.2026